

CompSci 234/NetSys 210
Advanced Topics in Networking
Winter 2019

**Unit 5: Multimedia
Networking Principle**

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Slide adopted from Profs. Ooi and Zimmerman, and Ross' materials

Agenda

- Media Streaming
- Video Quality Metrics
- Network Communication Models
- Different Transport Protocols
- Multimedia Friendly Internet

Networked Media Applications

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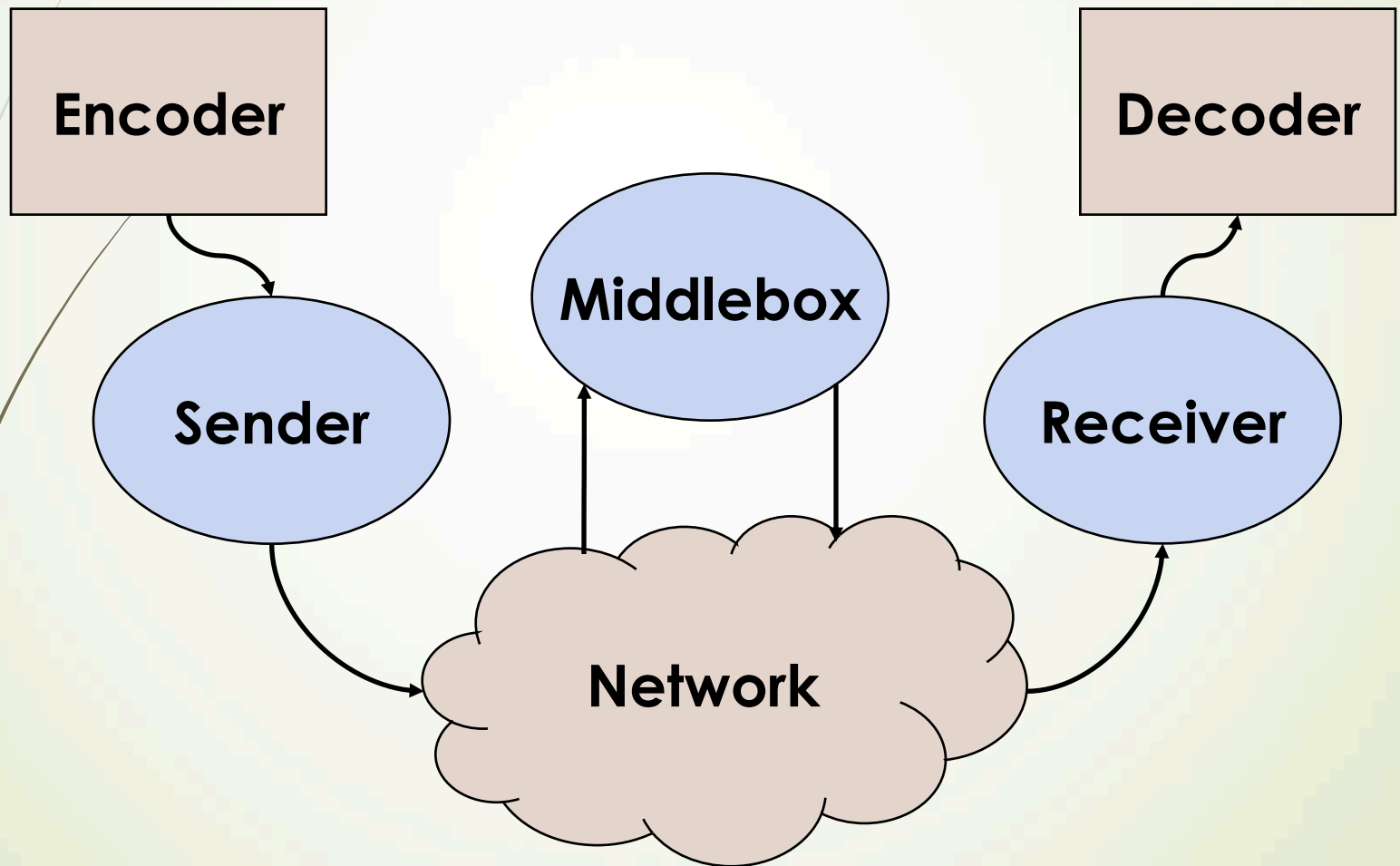
- Live Webcast and IPTV
 - Twitch, ...
- Pre-recorded Webcast
 - YouTube, ...
- Video Conferencing
 - Skype, ...
- Video on Demand
 - Netflix, ...

	Interactive	Non-Interactive
Live	Video Conferencing	Webcasts
Pre-recorded	Computer Games	Lecture/Video on Demand

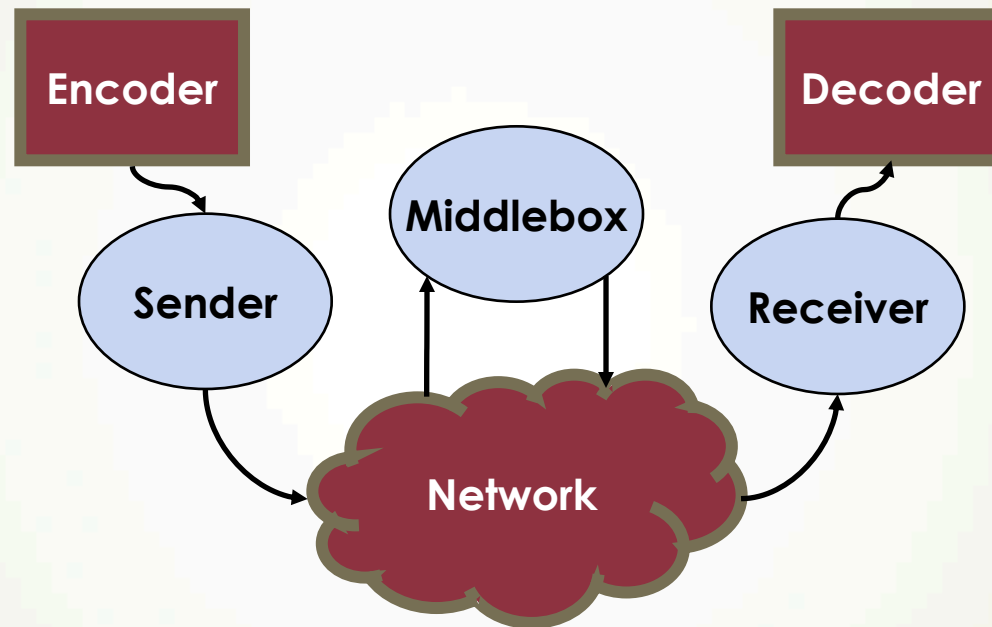
Unique Requirements of Media Streaming

- Plenty of bandwidth
 - Low latency
 - Bounded latency
 - Reliable networks
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- But, Internet was designed as a best-effort network

Bird's-Eye View of Media Streaming

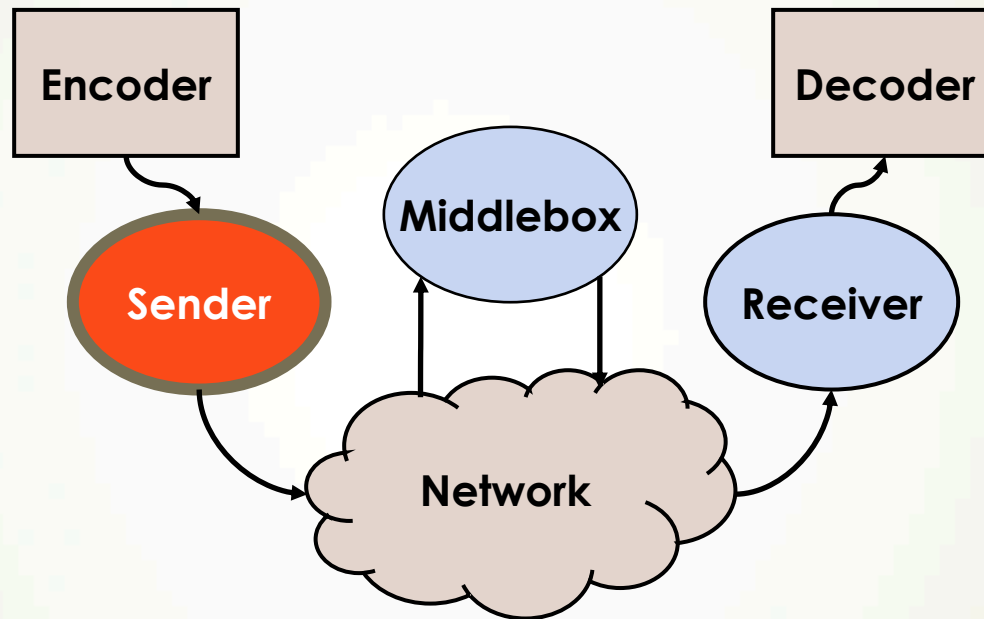


Challenges: Codec and Networking



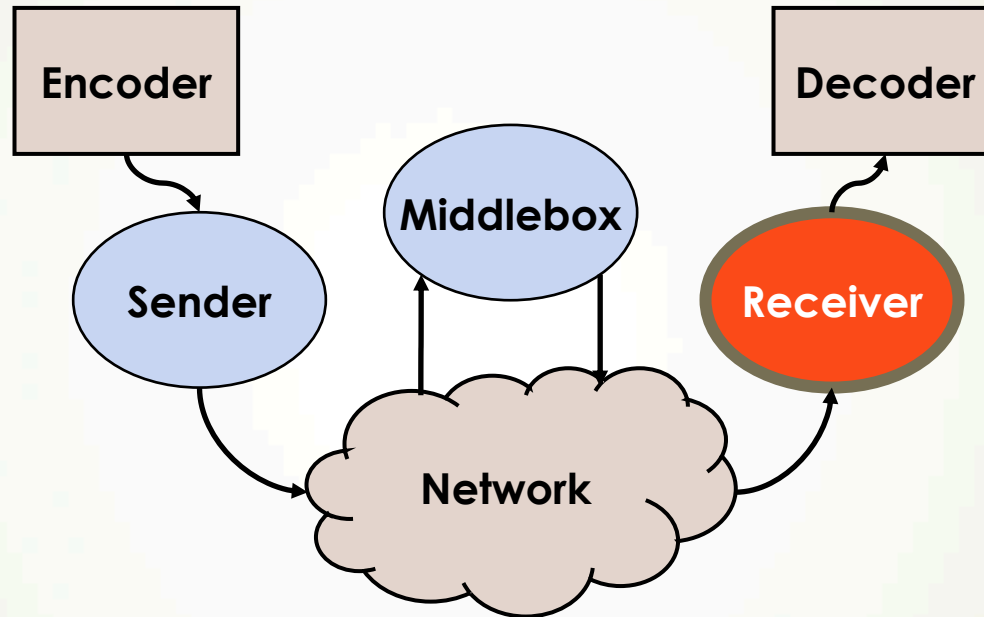
- Network Communication Model
- Media Compression

Challenges: Sender



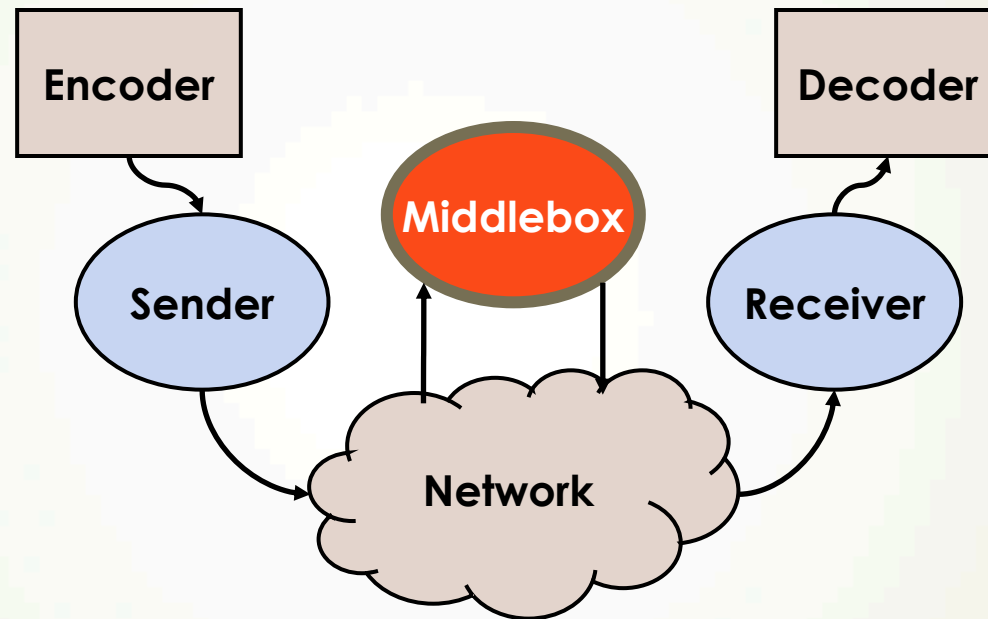
- Adaptation
- RTP conversational streaming
- Video on Demand, MPEG-DASH

Challenges: Receiver



- Buffering
- Audio/Video Synchronization
- Packet Loss Recovery

Challenges: Middlebox



- Caching
- Application-Level Multicast
- Media-Aware Network Elements

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How to Quantify Video (Media) Quality

- Basic concepts
 - Quality of Service (QoS)
 - Quality of Experience (QoE)
 - Resource reservation
- End-to-end path must respond to real-time requirements and provide a certain level of service quality

QoS vs QoE

- QoS – Quality of Service:
 - Network characteristics/behavior
 - Performance guarantees given by network provider based on measurements
- QoE – Quality of Experience:
 - Impact of network behavior on end user s
 - some imperfections may go unnoticed
 - some imperfections may render application useless
 - Not captured by network measurements
 - a 5% packet loss could be invisible if it affects background
 - a missed target due to a 100ms delay can affect game outcome

Possible QoE/QoS Factors

- Startup delay
- Stall events (number and duration)
- Visual quality measure:
 - PSNR: Peak Signal-to-Noise Ratio
 - SSIM, SSIMPlus Index: try to model human visual perception
- Quality switches

Sample Metric: Peak-Signal to Noise Ratio (PSNR)

- Mean Square Error (MSE):

$$MSE = \frac{\sum_{M,N} [I_1(m,n) - I_2(m,n)]^2}{M + N}$$

- M, N: Columns and rows, i.e., pixels

- PSNR:

$$PSNR = 10 \log_{10} \left(\frac{R^2}{MSE} \right)$$

- R: input value domain range, e.g., 255

Real-time Requirements (of Video)

- Periodic sampling: streams
 - Affects scheduling policy
- Fault tolerance
- Missed deadlines
 - Imperfect playout, e.g., frozen video
- Bandwidth demand
 - Bandwidth versus quality tradeoff

Service and Protocol Requirements

- Time-sensitive requirements
- High data throughput requirements
- Service guarantee requirements
- High or partial reliability requirements
- Cost-based fairness requirements

Resource Reservation

- Proper resource management helps to establish desired QoS/QoE (memory, bandwidth, CPU, ...)
- E.g.: network bandwidth: Circuit-switched versus Packet-switched
- Rule of Thumb

Shared resources can often be more (cost-) effectively used compared with dedicated resources

- E.g.: In circuit-switched telephone system “silence” will consume bandwidth

Reality on Offering Media Services with High-QoS/QoE

- (Networked) multimedia systems have certain requirements

But, we have

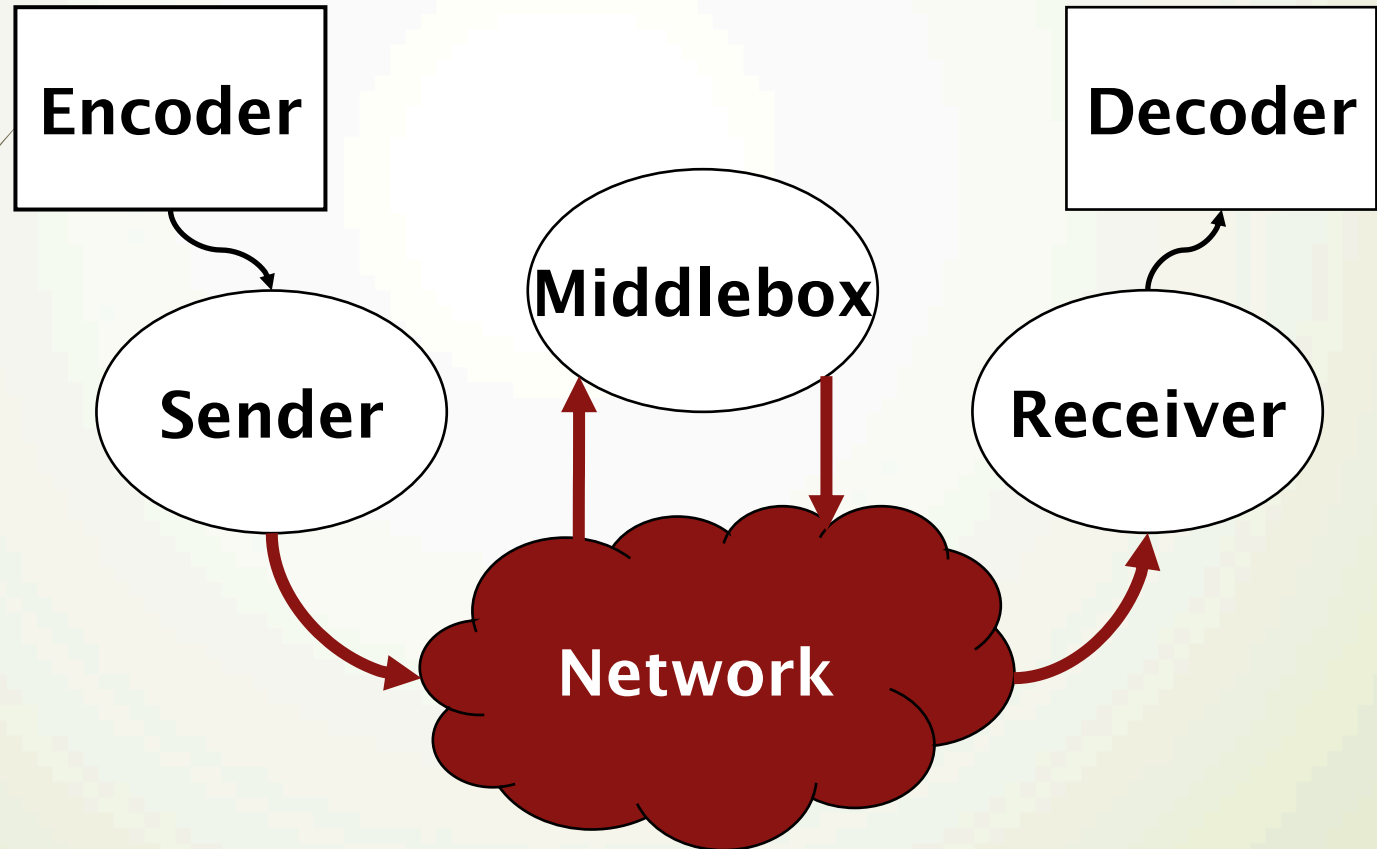
- Best-effort, shared network: Internet
- Non real-time OS: Windows, Linux

Need to find clever techniques

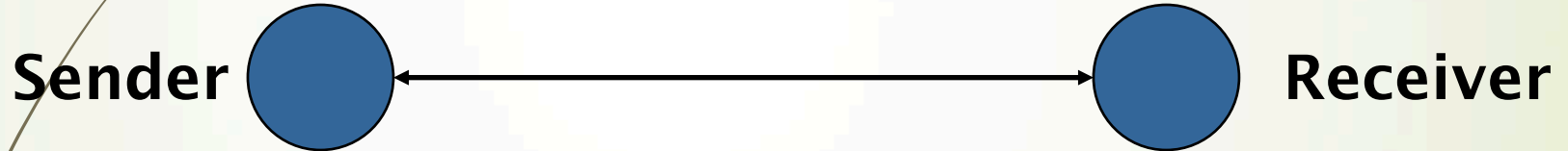
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Let's Talk About Networks (Network and Transport Layers)

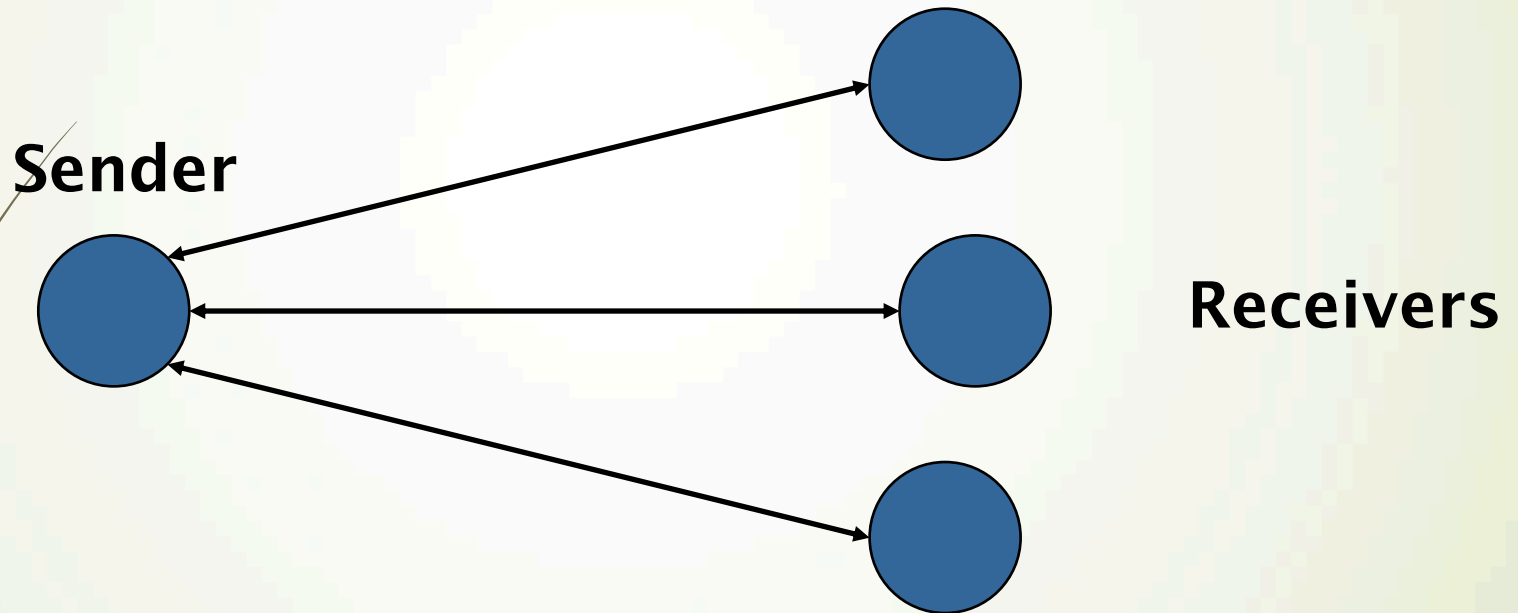


Comm. Model: Unicast



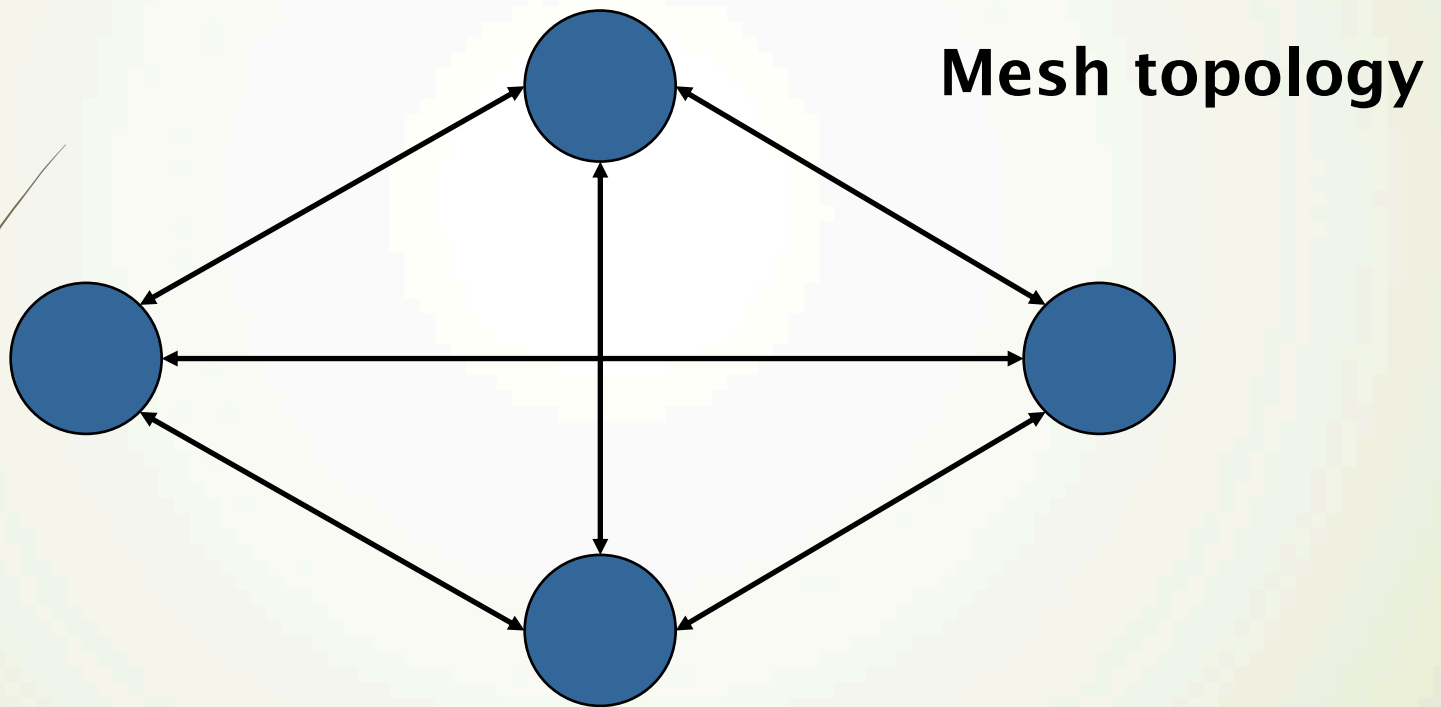
Traditional applications:
One-to-One

Comm. Model: Multicast (1/3)



Media applications:
One-to-Many

Comm. Model: Multicast (2/3)



Media applications:
Many-to-Many